24.0 FLUTED TUBE PANEL MODIFICATIONS

Initial tests of the 40 x 40 inch fluted tube panels resulted in the decision to add reinforcement members in an attempt to stabilize the tube walls and improve the strength of the panels.

24.1 Low Load Design

After the initial tests, the first low load fluted panel, 2A-1-P-1, was modified by inserting machined rods as spacers and identified as 2A-1-P-1M. The purpose of the spacers was to prevent flattening of the beads.

Figures 24-1 and 24-2 show views of the modified panel. Figure 24-2 also shows one of the machined spacers lying in the flute of one of the beads. A mastic material was applied to the rounded end and the shoulder on the spacers to hold the spacers in place until loaded.

24.2 High Load Design

After initial testing the first of the high load fluted panels, 2A-2-P-1 was modified in a manner identical to that of panel 2A-1-P-1.

A portion of the second high load fluted tube panel assembly, 2A-2-P-2M is shown in Figure 24-3. The figure shows inserts which were riveted to the first beaded face sheet. The inserts were subsequently attached to the second face sheet by blind fasteners.

Figure 24-4 shows a portion of one face sheet of panel 2A-2-P-3M with inserts bonded in place. A dense potting material was placed between the formed inserts and the beaded sheet in areas where the applied pressure had not resulted in good contact. Following the potting, five mechanical fasteners were installed in each insert, locating two in each bead wall, and one in the flute.